

Conditions for transmitting cryptochrome spin-state history to neural responses

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Abstract. Molecular memory can affect neural activity only if downstream processes can read it. Cryptochrome spin chemistry and redox-sensitive potassium channels motivate possible routes, but their quantitative connection remains unresolved. Here we show, using separate molecular, chemical and circuit models, that retaining a spin-state history does not ensure a persistent or consistently signed output. In a specified two-nucleus model, cycle-boundary populations reproduce the full density-matrix reaction yield, with a history-dependent yield difference of 0.00570. Reaction renewal introduces losses not explained by isolated storage. Chemical branches with equal final products lose their cumulative contrast, while rapid downstream writing requires an adequate substrate supply. In an illustrative 64-connection circuit, a 200 ms signal changes stopping probability by 14.16 percentage points under assumed high gain and favourable placement; the same route gives 0.09 points for a 12 ms signal. Lower gain and different placements reduce or reverse the effect. Reanalysis of published potassium-current data does not establish a predictive advantage for a dynamic redox model or identify the fast transduction gain. Extending earlier work on nuclear-population inheritance, these calculations distinguish storage, chemical throughput and timed neural readout. They specify conditional constraints and counterexamples, rather than a calibrated biological effect size. Measuring local chemical delivery, substrate turnover and synaptic gain in the same preparation is necessary to test whether these conditions coexist in vivo.

Keywords: cryptochrome, nuclear spin populations, redox memory, stochastic synaptic transmission, reaction kinetics

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1. Introduction

A molecular state can influence neural function only through a physical readout. The state must alter a reaction or interaction, generate a distinguishable downstream signal and reach a responsive cellular compartment while that signal remains relevant. These requirements separate the lifetime of a microscopic memory from its functional effect.

For a fluctuating biochemical input, the relevant quantity may be a transient flux, a bound cofactor population or a change in release probability, rather than a sustained change in bulk concentration. A useful quantitative account must therefore distinguish storage from writing, transport and detection.

Cryptochrome (CRY) provides an experimentally motivated setting for this problem. The radical-pair proposal connects electron–nuclear spin dynamics to spin-selective reaction yields [1, 2]. Magnetic sensitivity has been observed in purified European-robin CRY4 [3], but a molecular yield response does not by itself establish its transmission to a neuronal output. In *Drosophila*, blue-light responses of large ventral lateral neurons depend on CRY, and targeted CRY expression rescues neuronal light responses [4]. Hyperkinetic (Hk) was identified as a potassium-channel β subunit related to the aldo-keto reductase family [5]. In a mammalian Kv1.1–Kv β 1 preparation, oxidation of the bound cofactor modulates channel current [6]. In the fly, redox-sensitive Hk participates in CRY-dependent phototransduction: wild-type Hk, but not redox-impaired variants, restores the response in the corresponding rescue experiments [7]. Subsequent dCRY mutagenesis separated effects on acute membrane depolarisation from circadian resetting [8]. Together these findings connect a photoreceptor to neuronal excitability under light stimulation, while showing that its outputs need not share one mechanism. They do not measure an effect of a prepared nuclear-spin history in darkness, identify a CRY-sensitive presynaptic terminal or calibrate a behavioural stopping response.

A second line of evidence concerns chemical memory. In dorsal fan-shaped body neurons, lipid-peroxidation-related inputs and membrane-voltage manipulations alter potassium-current inactivation kinetics [9]. The persistence of a response after finite miniSOG illumination supports the interpretation that a channel-associated redox state retains a chemical history. This provides a reason to investigate Hk as a candidate storage element. It does not establish that the same input reaches Hk from dark CRY chemistry. Purified *Drosophila* CRY produces hydrogen peroxide under blue illumination, and CRY-expressing insect cells show a light-dependent reactive-oxygen-species indicator response [10]. Connecting these observations to lipid-derived substrates, Hk occupancy and synaptic release requires additional steps with their own stoichiometry, localisation and kinetics. The importance of production, transport and removal in local peroxide signalling is established in redox biology [11]; none of those processes is determined by a dimensionless radical-pair yield alone.

The microscopic history itself need not reside in long-lived coherence. Spin-selective chemistry can prepare unequal nuclear-state populations. Photoinduced nuclear polarisation has been reported in a flavoprotein system [12], in an artificial flavin–tryptophan dyad at Earth-field polarisation conditions [13], and in engineered LOV proteins [14]. These observations establish nuclear-population preparation in their respective systems, not the lifetime of the present protein-bound ^{14}N states. The wider flavoprotein literature distinguishes several mechanisms and reaction cycles [15]. Most directly, Wong and colleagues modelled nuclear-polarisation accumulation across

cryptochrome photocycles [16]. Their model removes coherences in the nuclear Zeeman basis before re-excitation and passes populations to the next radical-pair reaction. They found that the large improvements in magnetic-compass sensitivity seen in idealised cases did not persist under more realistic configurations and retention assumptions. This is a substantive warning against extrapolating favourable spin models to biology. Their observable was the directional anisotropy of a signalling-state yield; it differs from a history-dependent yield contrast at a fixed reaction setting.

At the neural level, transmitter release is stochastic and its probability can vary among individual synapses [17]. The placement and timing of a small perturbation can consequently be as important as its mean amplitude. A release-probability change at connections of opposite sign need not have the same effect. Likewise, improving detection of a stop cue and accelerating a stopping process after detection are different interventions. Independent-race theory [18] and interactive go-stop models [19] provide precedents for separating action initiation and cancellation. Stop-signal methodology also recognises that detection and trigger failures affect the interpretation of stopping measurements [20]. An illustrative circuit can separate these routes, but its gain and timing parameters must remain explicit assumptions until calibrated in a specified biological system.

However, the available observations do not determine whether a CRY spin-history contrast survives chemical convergence and finite substrate supply to produce a quantitatively identifiable neural effect. In this study, we examine this problem with separate models of nuclear-state inheritance, chemical routing, redox storage and stochastic synaptic integration, together with a reanalysis of published Hk-related electrophysiology. We find that populations reproduce the selected cycle-boundary yield, while reaction renewal, chemical convergence and circuit placement can attenuate or reverse the downstream contrast. The electrophysiological reanalysis constrains an observation model but does not identify the missing molecular-to-synaptic gain. The contribution is a set of explicit necessary conditions and counterexamples for transmitting history, with the uncalibrated interfaces retained in the interpretation.

2. Methods

2.1. Scope and interfaces

We analyse five numerical modules: a dark-reaction CRY spin model (M1), a candidate product-routing model (M2), a three-state Hk storage model (M3), a synthetic 64-connection circuit (M4), and a reanalysis of published current-inactivation data (M5). These are separate tests, not successive stages of an experimentally calibrated pathway. M4 uses the M1 yield contrast directly as a conditional input. It does not calculate that input through M2 and M3. Table 1 states the missing conversions. Numerical values below are either calculations in specified models, Monte Carlo estimates conditional on those models, or estimates from the stated reanalysis. Literature observations are

identified by citations.

Table 1. Each biological interface requires an independent calibration. The calculated modules do not close a single causal chain.

Interface	Quantities to connect	Missing calibration
Nuclear state to branch	Density matrix to yield per reaction	State preparation and recovery in vivo
Branch to local species	Yield to molecular flux or concentration	Event rate, stoichiometry, volume, delivery and removal
Species to Hk	Substrate concentration to writing rate and occupancy	Lipid chemistry, substrate supply and cofactor exchange
Hk to transmission	Occupancy to channel current and release probability	Localisation, voltage response and release coupling
Transmission to stopping	Release events to timed circuit outcomes	Biological connectivity, weights and delays

2.2. Full, population and reset models

The molecular state space retains two spin-1 ^{14}N nuclei, N5 and N10 of flavin, giving nine nuclear states. The full density-matrix model retains both diagonal populations and off-diagonal coherences. In a pure state $|\psi\rangle = \sum_j c_j |j\rangle$, population means $p_j = |c_j|^2$, not the complex amplitude c_j . Off-diagonal elements $\rho_{jk} = c_j c_k^*$ contain relative-phase information. The density-matrix definition also covers mixed states.

We fix the basis to the eigenstates of the closed-shell HQ nuclear Hamiltonian and define

$$\mathcal{D}_{\text{HQ}}(\rho) = \sum_j |j\rangle \langle j| \rho |j\rangle \langle j|. \quad (1)$$

For a completed reaction-cycle map $\mathcal{M}$, the full model propagates $\mathcal{M}$, whereas the population model propagates $\mathcal{D}_{\text{HQ}} \mathcal{M} \mathcal{D}_{\text{HQ}}$. The latter removes coherences at the cycle boundary while retaining unequal populations and diagonal two-nucleus correlations. A nuclear-reset control replaces the state by $I_9/9$. Both full and population models use the same quantum dynamics within the radical-pair reaction and the electron-containing escape state. Their comparison tests the need for coherence inherited across the specified boundary; it does not remove all quantum dynamics.

The reference condition uses a $50 \mu\text{T}$ field, a triplet-mixture initial electron state, a singlet reaction rate of 10^7 s^{-1} and an escape rate of $4 \times 10^6 \text{ s}^{-1}$. Reduction, HQ waiting and product-branch recovery rates are each 10 s^{-1} ; escape-state oxidation is 4 s^{-1} . Electron exchange and dipolar couplings are set to zero. After 20 completed preparation cycles, the chemical state is aligned to HQ and the next product-branch yield is evaluated. A completed-cycle count is not a fixed elapsed time or a claim of steady state.

The reaction maps use a Haberkorn-type spin-selective loss term [21, 22], and the waiting-state noise is represented by a positive-coefficient Lindblad dissipator [23]. These are modelling choices: reaction theory can generate additional reactive exchange and dephasing terms [22]. The generators use previously computed tensors and a reduced noise model derived from a short oxidised-CRY molecular-dynamics trajectory. The adopted trajectory spans 6 ps and the noise construction uses a 1 ps window. State transfer of these inputs, fixed overall orientation and extrapolation of the generator to long times are model assumptions. Supplementary Methods S1 specifies the cycle construction and archived operators. The resulting relaxation times are not measured biological nuclear lifetimes. Earlier molecular-dynamics/spin-dynamics work evaluated electron relaxation in cryptochrome using Bloch–Redfield theory [24]; it motivates explicit treatment of fluctuations but does not calibrate our closed-shell nuclear generator.

2.3. Writing, storage and reading nuclear populations

Let C insert populations as diagonal density matrices in the HQ basis and let D be the dissipative generator. The projected population generator is $G = C^\dagger DC$. On the normalisation-preserving subspace, we write the prepared deviation as $\delta p = \sum_{k=1}^8 a_k v_k$. If b_k is the next-reaction sensitivity to mode v_k , the population approximation to the yield contrast during isolated HQ storage is

$$\Delta Y(t) = \sum_{k=1}^8 a_k b_k \exp(-\lambda_k t). \quad (2)$$

The functional contribution of a mode depends on both its preparation and its readout, not only on its relaxation rate λ_k . To distinguish storage from renewal, we also apply a common subsequent reaction map to two differently prepared states and track their difference. This measures survival of the pre-existing contrast rather than newly written history. Local populations and diagonal correlations are separated without interpreting the latter as entanglement.

2.4. Chemical routing and substrate-limited storage

M2 assigns candidate peroxide-forming and escape branches, denoted P and E. This assignment is a chemical modelling assumption, not a measured dark-CRY stoichiometry inferred from the light-driven experiments in [10]. Fast superoxide spin relaxation places restrictive conditions on proposed flavin–superoxide magnetic sensors [25]. Simulations of transient superoxide binding to avian CRY4 identify possible immobilisation sites [26], but do not validate the zero additional oxygen-relaxation reference used here. We consider the limit in which E fully reoxidises, superoxide fully dismutates, and products from both branches reach the same compartment. With P probability q and E reoxidation rate k_E , the mean cumulative peroxide yield per HQ reaction is

$$Y_{\text{H}_2\text{O}_2}(t; q) = 1 - \frac{1-q}{2} \exp(-k_E t), \quad \Delta Y_{\text{H}_2\text{O}_2}(t) = \frac{\Delta q}{2} \exp(-k_E t). \quad (3)$$

The half-equivalents describe ensemble means. Removal and branch-specific delivery are considered separately. No direct peroxide-to-4-ONE reaction or fixed conversion efficiency is supplied.

Motivated by the redox interpretation of Hk experiments [7, 9], M3 represents reduced-cofactor-bound, oxidised-cofactor-bound and vacant sites by fractions z_R, z_O, z_U . These labels are distinct from the CRY chemical states. The assumed kinetic scheme is

$$\begin{aligned}\dot{z}_R &= -a(t)z_R + hz_U, \\ \dot{z}_O &= a(t)z_R - bz_O + oz_U, \\ \dot{z}_U &= bz_O - (h + o)z_U, \quad z_R + z_O + z_U = 1.\end{aligned}\tag{4}$$

Writing is $R \rightarrow O$, oxidised-cofactor departure is $O \rightarrow U$, and rebinding is $U \rightarrow R$ or $U \rightarrow O$. For the finite-input calculation, $a(t) = a_0 \exp(-t/\tau)$, the initial state is all R, and $o = 0$. The illustrative values $a_0 = 10 \text{ s}^{-1}$, $b = 0.01 \text{ s}^{-1}$ and $h = 100 \text{ s}^{-1}$ are not fitted neuronal rates. We separately calculate the total response and the response to a 1% increase in input amplitude. This 1% perturbation is a sensitivity probe, not an inferred conversion of Δq .

For a concentration C_s^{chem} in μM and volume V in μm^3 , the mean substrate count is $602.214076 C_s^{\text{chem}} V$. With one consumed substrate per write and no resupply or reuse, the number of new oxidised sites cannot exceed the initial substrate count. This conservation constraint is distinct from the supplied-bath approximation in equation (4).

2.5. Synthetic transmission and stopping circuit

M4 maps the molecular yield to a cue-aligned binomial preparation with N slots. This is a conditional probe protocol, not a model of a natural cue preparing 20 molecular cycles. The two arms have preparation probabilities q_0 and q_1 and common subsequent recovery towards q_0 . At each 10 ms packet, the release probability at connection j is

$$p_j = p_0 + \alpha_j C_p \tanh \left[\frac{g(n/N - q_0)}{C_p} \right], \quad C_p = 0.95 \min(p_0, 1 - p_0)k, \tag{5}$$

where $p_0 = 0.45$, $\sum_j \alpha_j = 1$, k is the selected-connection count and g is an assumed dimensionless gain. Bernoulli release is an idealisation of stochastic synaptic transmission [17]; the specific saturating transfer function and its gain are assumptions of this model. The same nonlinear map transmits both the intervention and baseline chemical fluctuations. Matching $\sum_j \Delta p_j$ across allocations does not match weighted input or its variance.

The 64-connection model distinguishes a go process, a cue-trigger process and a stopping process that becomes active after triggering. This architecture is inspired by the race-model literature [18, 19, 20]; its packet rules are not a fit to those experiments. It compares equal-weight voting, signed weighting, 40 ms temporal integration and a mutually inhibitory competition model. The top- k allocation selects connections before evaluation by $w_j \exp[-d_j/(40 \text{ ms})]$. Other allocations are diffuse,

random, weak and opposite-sign. This selection favours the adopted readout; it is not identification of biological CRY-sensitive synapses. Further equations and seeds are given in Supplementary Methods S4.

The main condition uses $N = 10\,000$, $k = 8$, $g = 1024$, and chemical lifetimes of 12 or 200 ms. The cue occurs at 200 ms, chemical arrival follows after 10 ms, and connection delays range from 0 to 30 ms. Earliest detection is 40 ms after the cue and earliest stopping is 70 ms after detection. These are synthetic timing choices. A trial is stopped if the stopping process finishes before go onset; continuation and unresolved outcomes are recorded separately over 1 s. This construction is not a membrane-equation or electroencephalographic observation model.

2.6. Reanalysis of published potassium-current data

M5 uses the public source data for figures 4(f) and 6(d) of Rorsman *et al.* [9]: current-inactivation time constants from 33 wild-type adult dorsal fan-shaped body neurons with nominal intracellular 50 μM 4-ONE. We fit 17 cells and hold out 16 using odd/even within-figure cell identifiers. All 40 min observations are excluded from fitting. Fast and slow components and repeated observations of each cell remain together. The observation ledger retains 38 missing values among 378 scheduled entries; entries are not independent cells.

The effective dynamic model predicts log time-constant changes from a shared latent coordinate with rest and voltage-train-dependent rates. It has five fitted parameters. The static comparator has two fitted post-baseline levels, one per component; a no-voltage model is also fitted. Predictions are conditional on each observed baseline, rather than predictions of an unseen baseline. Bounded multistart least squares weights cells and components equally after standardisation by training-data scales. Supplementary Methods S5 gives the equations, split, search bounds and bootstrap procedure. This minute-scale analysis does not fit the separate millisecond-input rates in equation (4) or the gain in equation (5).

2.7. Numerical uncertainty and conditional output bounds

The six main circuit conditions use three seeds with 8192 paired trials each, totalling 24 576 pairs per condition. Common random numbers couple the full and reset arms, following the paired-simulation approach [27]. Variance reduction is not presumed for every nonlinear output; the uncertainty is calculated from the actual paired differences. Reported circuit intervals are 95% Monte Carlo intervals calculated from paired differences; they exclude biological parameter uncertainty. The lower-resolution gain grid uses 3072 pairs per condition. Existing checks include a separate molecular resolvent calculation, alternative ODE integration, packet-probability enumeration and reclassification of stored circuit event times. Preparation of this article reuses these simulations and independently checks selected numerical summaries; it does not constitute a new biological experiment.

Using the standard coupling and total-variation characterisations of probability differences [28], a gain-independent bound follows for the specified one-cue preparation. Set $\delta = |q_1 - q_0|$. Under a monotone coupling of the N Bernoulli slots, the probability that any slot differs is $1 - (1 - \delta)^N$. When equal prepared counts imply equal subsequent trajectories and outcomes under common updates, the stopping indicators satisfy

$$|P(S_1 = 1) - P(S_0 = 1)| \leq P(S_1 \neq S_0) \leq 1 - (1 - \delta)^N. \quad (6)$$

This requires a single intervention-dependent preparation, common downstream rules and equality-preserving coupling. Repeated intervention-dependent injections require another analysis. If the sole informative input is instead one count with a common conditional readout, total variation between the two binomial count distributions supplies a tighter bound. That marginal bound does not automatically apply to M4, which reads correlated background history before chemical arrival. The proof and a counterexample are given in Supplementary Methods S6; neither bound is claimed as a new general theorem.

3. Results

3.1. Population readout reproduces the selected yield, but renewal erases stored history

After 20 preparation cycles, the full and reset yields are 0.244088539902 and 0.238386607449, respectively, giving $\Delta q = 0.005701932453$. This is an absolute reaction-yield difference of 0.5702 percentage points, not a stopping-probability difference. The full-population yield discrepancy is 9.4×10^{-12} , below the 2.7×10^{-10} discrepancy in the independent solver diagnostic. The tiny residual therefore does not resolve a coherence contribution.

The eight non-equilibrium modes contribute unequally (figure 1). Diagonal correlation modes that preserve both one-nucleus marginals account for 43.41% of the selected readout contrast. This is a mode-decomposition share, not a biological quantum fraction. The longest-lived mode is not the largest contributor. Local modes mainly describe rank-2 population alignment, so a single magnetisation relaxation time T_1 does not represent the whole readout.

Isolated HQ storage gives half-decay and $1/e$ times of 1.635 and 2.454 s. Applying one additional completed reaction cycle instead retains 40.09% of the pre-existing yield contrast. Removing HQ dissipation changes this to 41.48%, whereas removing escape-state dissipation gives 88.75%. Thus population relaxation describes the isolated storage test, but reaction-state evolution and renewal also erase history. The seconds-scale estimates extrapolate the adopted short-trajectory noise model and cannot be interpreted as measured neuronal memory times.

The stored one-factor scan contains 72 settings and three models per setting. The history contrast changes sign at a P-branch recovery rate of 0.1 s^{-1} (-0.00108379), at escape-state oxidation of 10^4 s^{-1} (-0.00103585), and with a 100-fold closed-shell noise multiplier (-2.85556×10^{-5}). Slowing HQ waiting to 0.1 s^{-1} reduces it to $+0.00072760$.

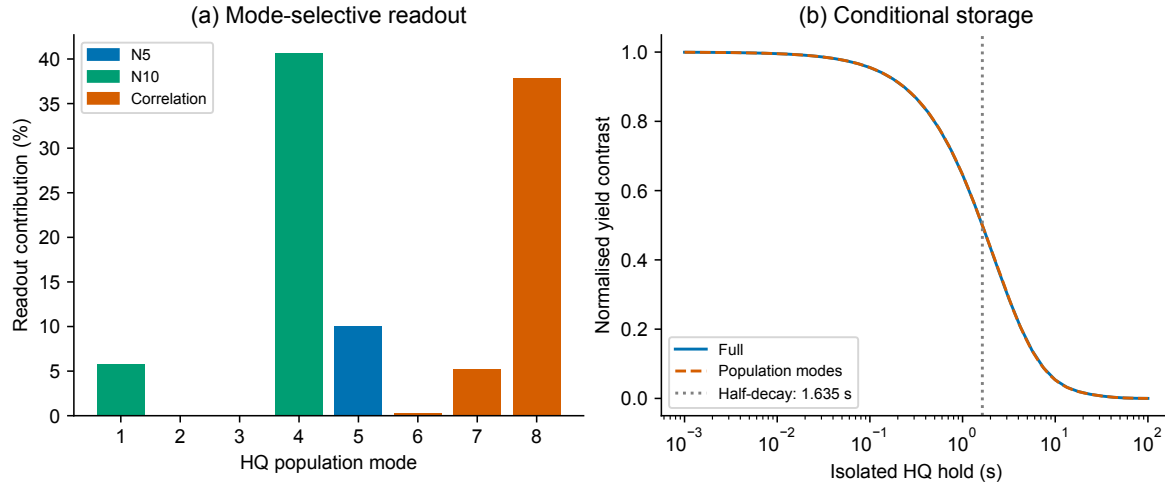


Figure 1. Writing and readout determine which nuclear-population modes matter. (a) Contributions of the eight modes to the prepared yield contrast. Local and correlation modes are distinguished; their contributions sum to the population-model contrast. (b) Readable contrast during isolated HQ storage, normalised to its initial value, from the full calculation and population mode sum. The marked half-decay time is a conditional model result. It is not an in-vivo nuclear lifetime.

These scans are assumed ranges, not samples from a physiological parameter distribution. Short phase-preserving waits and near-degenerate counterexamples also break the population approximation (Supplementary Methods S2). The reference agreement therefore establishes neither an exact structural equivalence nor general validity over orientations, couplings and nuclear sets.

3.2. Chemical convergence and supply constrain downstream memory

Under the equal-endpoint assumptions, both P and E ultimately yield one peroxide equivalent per HQ reaction. Equation (3) then gives a transient contrast that vanishes at long times (figure 2(a)). With assumed reoxidation and removal rates of 1 and 5 s⁻¹, the retained compartmental difference falls from +0.002851 equivalents per HQ event immediately after the reaction to +0.0007275 at 200 ms, and becomes -0.0000963 at 2 s. Removal can thus reverse the residual contrast even though cumulative production initially favours the higher-P arm. Unequal delivery or a transient-sensitive downstream process can retain information, but total final product alone cannot in this example.

In the independently supplied Hk model, exponentially decaying inputs of 12, 50 and 200 ms produce peak oxidised fractions of 11.30%, 39.20% and 85.67%, respectively (figure 2(b)). The 200 ms input peaks near 1.024 s. These values are responses to the whole chemical input. The separate 1% amplitude perturbation gives an oxidised-fraction difference of only 0.3546 percentage points at 200 ms for the 200 ms input. Neither quantity is a calibrated CRY-history effect.

Substrate accounting gives a further constraint. A closed 0.01 μm^3 compartment

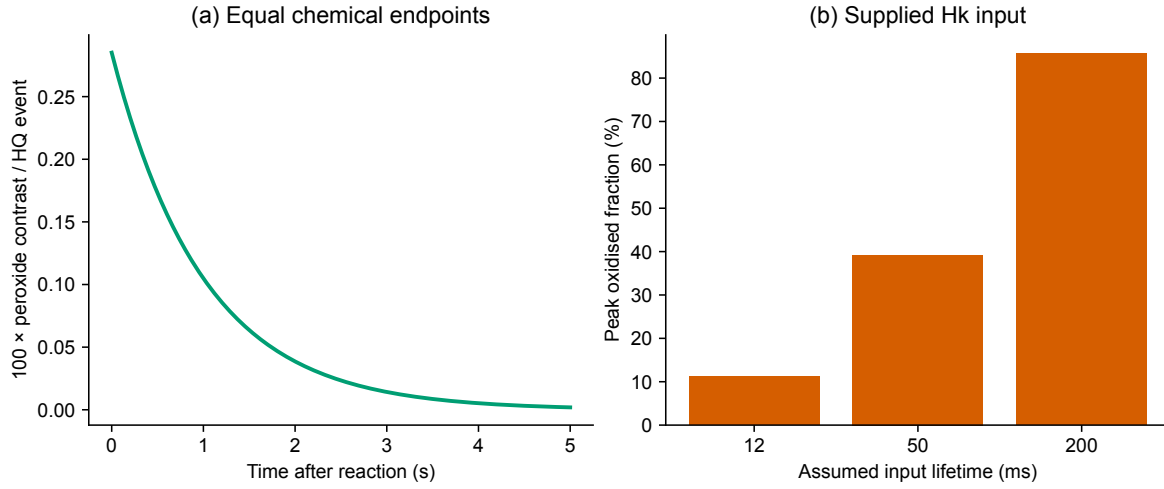


Figure 2. Chemical storage requires a readable transient and sufficient substrate. (a) Cumulative peroxide contrast in the equal-endpoint model, $\Delta q \exp(-t)/2$, expressed as 100 times the mean molecular yield per HQ event. (b) Peak oxidised Hk fraction under independent exponential writing inputs, with $a_0 = 10$, $b = 0.01$ and $h = 100 \text{ s}^{-1}$. These panels are separate conditional models. The Hk response is not driven by the peroxide trace through a calibrated conversion.

at $1 \mu\text{M}$ contains 6.02 substrate molecules on average. Oxidising 85.67% of 200 initially reduced sites requires at least 171.34 writes on average. Without resupply or reuse, the initial mean substrate count can increase mean occupancy by no more than 3.01%. Fast writing and slow erasure are therefore insufficient without a compatible material supply. The closed-compartment bound and supplied-bath peak describe different conditions, rather than contradictory calculations.

3.3. Target, timing and gain determine the synthetic circuit effect

In the top-eight, temporal-integration, $N = 10\,000$, $g = 1024$ condition, the 12 ms signal changes stopping probability by -0.195 ± 0.055 , $+3.601 \pm 0.240$ and $+0.090 \pm 0.037$ percentage points when applied to go, trigger and post-trigger stop, respectively. For 200 ms, the corresponding changes are -15.157 ± 0.448 , $+18.599 \pm 0.494$ and $+14.160 \pm 0.436$ points (figure 3). The uncertainties are paired Monte Carlo interval half-widths. A short-lived input can affect early detection while leaving little contrast for later stopping.

For the 200 ms stop intervention, diffuse allocation over all 64 connections gives approximately +2.65 points, top-eight allocation gives +14.16 points and opposite-sign allocation gives -15.25 points at matched total probability modulation. This does not isolate a benefit of sparsity: weighted mean input and variance differ. In a single-packet equal-weight majority rule, relabelling an otherwise identical set of k modulated connections as top, weak or opposite cannot change the vote probability. Weights and timing must be specified before assigning functional importance to a small synaptic subset.

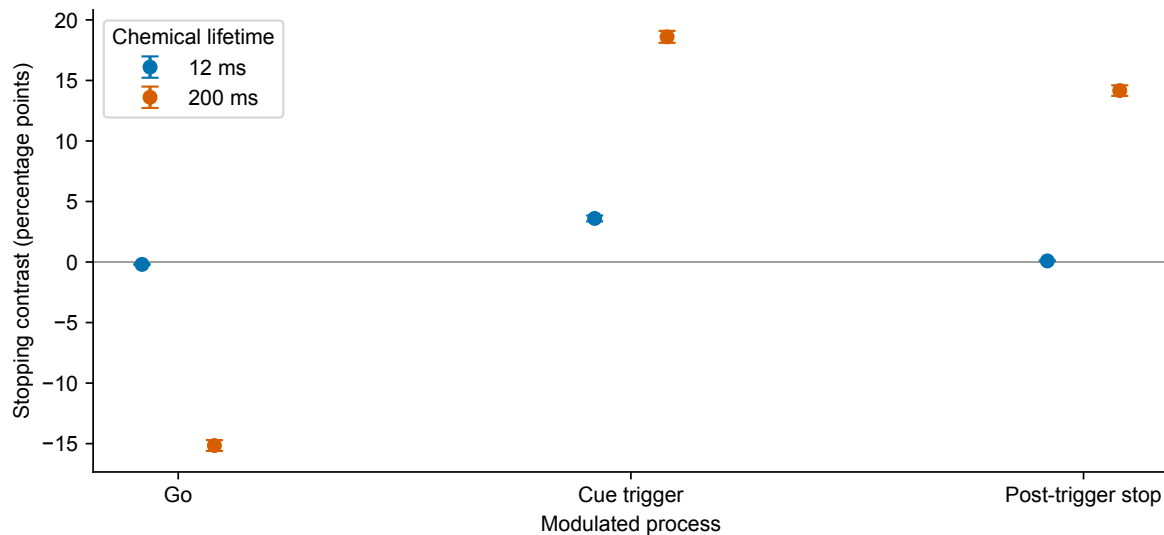


Figure 3. The same assumed molecular input changes stopping differently across circuit targets and lifetimes. Changes in stopping probability for go, trigger and post-trigger stop modulation, with top-eight allocation, temporal integration, $N = 10\,000$ and $g = 1024$. Each point represents 24 576 paired trials. Error bars are 95% Monte Carlo intervals; they do not represent uncertainty in biological gain, connectivity or chemistry.

The saved gain grid gives 0, 4.720, 11.361 and 14.974 points for $g = 0, 64, 256, 1024$, respectively, with 3072 paired trials per point (figure 4(b)). The $g = 64$ interval is 3.970–5.470 points. The high-gain grid estimate differs from the 24 576-pair confirmation estimate and is not another biological condition. A second network gives 14.225 points at high gain, but two structures do not characterise network-population uncertainty. No verified extended high-gain curve is used here.

When the maintained chemical background is removed and both pools decay towards zero, the 200 ms top-eight stop difference becomes zero and the reference stopping probability also becomes zero. This is a change in operating point, not a pure lifetime comparison. At top-eight allocation only the selected connections approach the lower release bound; the remaining 56 retain baseline release probability. The result identifies another assumption supporting the large conditional effect.

3.4. Published current data do not determine fast transduction gain

The dynamic and static models give standardised root-mean-square errors of 0.761 and 0.956 on the 17 training cells, but 0.839 and 0.806 on the 16 held-out cells. On the seven held-out cells with 40 min observations, the corresponding errors are 1.327 and 1.323 (figure 5). The mean cell-level dynamic-minus-static squared-error difference is +0.05443, with a stored cell-bootstrap interval of $[-0.16612, +0.25133]$. At 40 min it is +0.00925, with interval $[-0.01231, +0.02953]$. These intervals establish neither dynamic superiority nor equivalence or the biological truth of the static comparator.

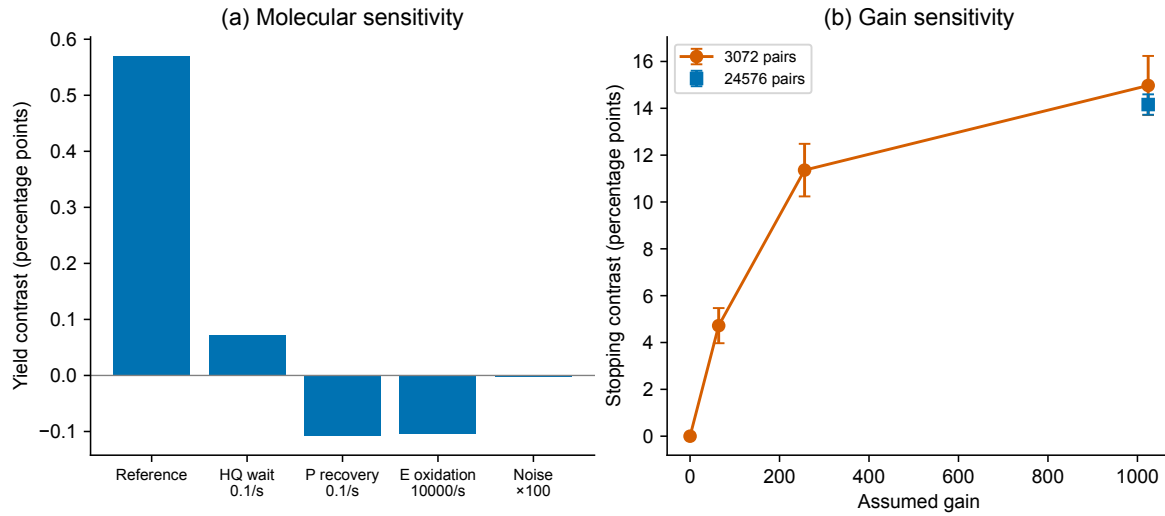


Figure 4. The molecular contrast and circuit effect depend on assumed parameters. (a) Selected one-factor changes from the molecular reference condition show attenuation and sign reversal. Recovery and oxidation rates are in s^{-1} ; the closed-shell noise multiplier is dimensionless. (b) The saved circuit gain scan uses 3072 paired trials per point; the separate confirmation uses 24 576 pairs. Error bars are 95% Monte Carlo intervals. The gain has no physiological calibration.

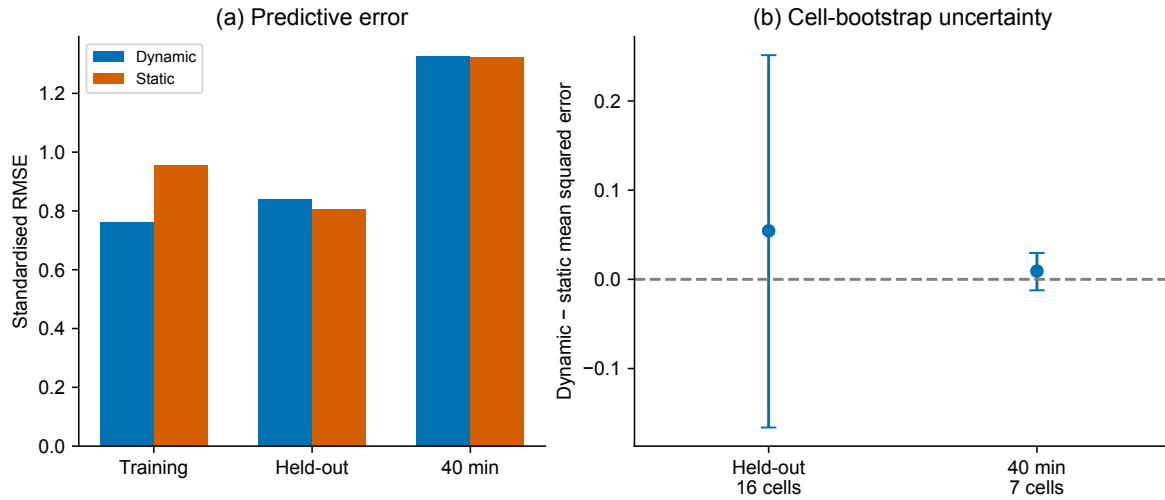


Figure 5. The reanalysis does not establish a held-out predictive advantage for the dynamic model. (a) Standardised prediction errors for training cells, held-out cells and the held-out 40 min condition. (b) Cell-averaged dynamic-minus-static squared-error differences and stored 95% cell-bootstrap percentile intervals; positive values favour the static comparator. These intervals condition on the fitted observation framework and observed baselines. Animal and batch dependence cannot be assessed from the available identifiers.

Occupancy and observation gain remain confounded: changing the latent occupancy scale and inversely changing the observation gain can preserve the predicted time constants. The fitted slow-component excursion reaches the upper search bound of

4 in 12 of 200 training-cell bootstrap fits. Expanding that bound to 8 places its 97.5th percentile at approximately 7.99998; the upper uncertainty remains unstable, although the principal fitted prediction is unchanged. Sustained-input fits also do not uniquely determine erasure after input removal. These are limitations of this reanalysis, not refutations of the original finite-input redox-memory observations.

3.5. Finite preparation constrains output differences conditionally

For $\delta = 0.005701932453$, equation (6) gives upper bounds of 0.5702, 3.3728, 13.8150 and 14.3065 percentage points for $N = 1, 6, 26, 27$. A true probability difference of 14.1602 points therefore requires at least 27 slots under this preparation and coupling, but 27 slots are not sufficient. For the main $N = 10\,000$ case, the bound is nearly 100 points and supplies no useful quantitative restriction. The slots are not measured Hk sites or the substrate molecules in the preceding compartment example.

Under the additional single-count common-readout assumptions, the binomial total-variation bounds are 5.3089 and 49.4844 points for $N = 100$ and $N = 10\,000$. Applying these values to the complete history-dependent circuit would omit informative correlations with the background trajectory. The distinction identifies what must be specified before using an information bound to constrain a neural output.

4. Discussion

4.1. A classically representable memory can follow quantum reaction dynamics

The close full-population agreement supports a restricted statement: the selected yield does not require coherence to cross the specified HQ boundary within numerical resolution. The retained state is a population distribution, including correlations, written and read by the adopted spin-selective reaction. Because both models share that reaction, their agreement does not demonstrate an independently established classical chemical mechanism or exclude all quantum contributions. Replaying the same chemical trajectory into the same classical downstream circuit likewise reproduces its output without identifying the trajectory’s microscopic origin.

Wong *et al.* [16] are a direct precedent for population inheritance. The distinction here concerns a specified dark-reaction cycle, HQ-basis reduction, write-read mode decomposition and separation of isolated storage from reactive renewal. It is not the first proposal that nuclear populations affect a subsequent yield. Their warning about realistic spin systems is relevant: fixed orientation, two retained nuclei and zero electron exchange/dipolar couplings are substantial restrictions in the present reference calculation. A matched-limit numerical comparison and an error indicator tested on withheld physical conditions remain necessary to establish the generality of the reduction.

The study also addresses a different question from quantum reservoir computing, where driven quantum dynamics supply features to a trained readout [29]. A related

preprint by Wakaura and Tanimae examines cryptochrome radical pairs in that framework and reports a restricted biological readout and a microsecond-scale memory horizon [30]. That result concerns its radical-pair reservoir protocol. The present calculation follows nuclear populations between reactions and does not evaluate a trained-readout capacity or establish quantum computational advantage. Its longer conditional storage time therefore does not by itself overturn the earlier reservoir limitation.

The negative result of [16] should also retain its scope. Section III D reports a 50–100% relative anisotropy enhancement in four-nucleus, S-branch-retention cases with realistic radical configurations, while the paper concludes that the absolute compass-performance improvements are unlikely to be substantial. This is neither complete disappearance of population history nor evidence for a large neural effect. Supplementary Methods S7 contrasts the observables and assumptions directly.

4.2. Long lifetime is not enough to determine biological impact

The calculations expose different failure modes along a possible transduction route. Chemical convergence removes a final total-yield contrast. Finite supply limits the amount that can be written into a long-lived state. Circuit sign and timing determine whether a transmitted contrast promotes stopping, continuation or neither. These constraints cannot be collapsed into one memory time or one amplification factor. In particular, the 85.67% total Hk oxidation and the 14.16-point circuit contrast are outputs of different supplied-input assumptions and are not sequentially calibrated predictions.

Other experiments illustrate the distinction between microscopic response and persistent output. Chandler and colleagues observed light-induced structural switching and dark recovery in purified dCRY, but did not detect a magnetic-field effect on the structural readout under the tested conditions [31]. Conversely, optically detected spin resonance in algal cryptochrome and engineered iLOV provides a controlled spin-chemistry readout; the slow iLOV signal buildup was attributed to a shift in chemical-state populations [32]. Neither observation measures persistent nuclear coherence or a native stopping effect.

The empirical motivation for Hk storage survives the lack of dynamic superiority in M5. The finite illumination and voltage-dependent recovery experiments in [9] provide evidence distinct from the sustained-input subset fitted here. CRY-dependent neuronal phototransduction [4, 7], light-driven CRY reactive-oxygen production [10], and Hk-associated memory [9] jointly motivate candidate steps. They do not constitute independent replications of a single dark-CRY-to-synapse pathway. A claim that the resulting brain effect is sufficiently large would require calibration of those interfaces in a common preparation.

CRY also has routes to neural and behavioural effects that do not require the proposed spin-history mechanism. In the mouse nucleus accumbens, CRY perturbation

affects D1-related activation and behaviour, and heterologous-cell assays support regulation of D1/Gs signalling [33]. A Cry1 exon-11 deletion study reports altered excitability of putative D1 neurons and loss of CRY1–Gs inhibition in a separate cell assay [34]. In the fly antenna, interventions support a light-sensitive support-cell CRY–ROS–TRPA1 route affecting olfactory responses [35]. These studies identify alternative mechanisms and compartments that a spin-specific test must distinguish; they do not calibrate the present dark-reaction pathway.

The operational circuit veto means stopping before motor onset. It does not model subjective intention, free will or a conscious decision at an individual synapse. Readiness potentials are a possible future application, but require a population-to-electroencephalogram observation model and movement-aligned trial analysis. Brain–computer-interface experiments have shown cancellation of some self-initiated movements after readiness-potential onset [36]; they do not locate the cancellation decision at individual synapses. Accumulator models can generate readiness-potential-like averages [37], while alternative explanations and the status of single-trial fluctuations remain debated [38]. Neither line of work identifies CRY as the source of those fluctuations. This motivates careful causal tests rather than a molecular attribution based on an averaged waveform.

4.3. Measurements that would distinguish the candidate mechanisms

A useful experimental comparison would separate total reaction output from its local time course. Matching final product while varying delivery timing would test a transient readout; measuring local substrate together with channel response after input removal would separate substrate persistence from storage in the channel-associated state. Varying reaction count at matched elapsed time and chemical occupancy would test renewal loss against isolated nuclear relaxation. These are proposed discriminating interventions, not experiments performed here.

Circuit tests additionally require identifying the affected connections and measuring their release response. To attribute an advantage specifically to sparse placement, comparison groups should match weighted mean input and relevant covariance, with selection criteria independent of the chosen readout. A quantitative Hk experiment-design analysis should propagate occupancy-scale and kinetic uncertainty into predictions for washout, voltage changes and reapplication, then compare model separation with measurement noise. The present sustained-input fit alone does not supply that separation.

The most consequential limitations are therefore biological calibration and structural generality. The molecular generator extrapolates short-trajectory, state-transferred inputs; the chemistry uses candidate branches and idealised supply; the circuit uses assumed gain, selected placement and two network structures; the current reanalysis conditions on measured baselines and lacks animal/batch identifiers. Monte Carlo precision does not reduce these uncertainties. The modules quantify individual

constraints, but their simultaneous satisfaction in vivo remains untested.

5. Conclusion

The specified cryptochrome model stores a readable reaction history in cycle-boundary nuclear populations and correlations, while subsequent reaction cycles introduce losses beyond isolated HQ relaxation. Downstream models show that equal chemical endpoints, limited substrate and delayed or oppositely signed circuit targets can suppress or reverse the output contrast. These results extend the analysis of population inheritance to explicit readout and transduction constraints without determining a native neural effect size. The practical implication is that a long memory time, a large total chemical response and a large conditional circuit effect cannot substitute for the intervening calibrations. Measurements linking local chemical delivery, cofactor turnover and synaptic release in the same preparation are needed to turn these conditional results into a test of biological action.

Data availability

Numerical source tables, frozen molecular operators, selected paired circuit event records and analysis code accompany this manuscript as Supplementary Data. The electrophysiological source observations are available with Rorsman *et al.* [9]; the derived observation ledger records source-sheet and cell locations. The accompanying reproduction starts from frozen molecular operators and saved simulation outputs; it does not regenerate the upstream molecular-dynamics trajectory. No public repository identifier has been assigned to the present computational package.

Use of AI-assisted tools

OpenAI Codex and Claude were used during computational development, internal review and manuscript preparation, including English drafting. Their outputs were assessed against source documents, implementation records and numerical checks. No new biological observations were generated by these tools. AI-generated assessments are not treated as independent experimental evidence.

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